# MIT 8.012 Problem Set 4 - Problem 3.15 (K&K)

3.15 A rope of mass $M$ and length $l$ lies on a frictionless table, with a short portion, $l_0$, hanging through a hole. Initially the rope is at rest.

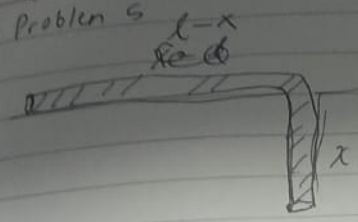
*a.* Find a general equation for $x(t)$, the length of rope through the hole.

*Ans.* $x = Ae^{\gamma t} + Be^{-\gamma t}$, $\gamma^2 = g/l$

*b.* Evaluate the constants $A$ and $B$ so that the initial conditions are satisfied.

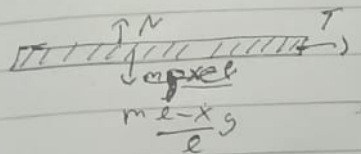
Solution (handwritten) follows on the next pages.

Problem 5



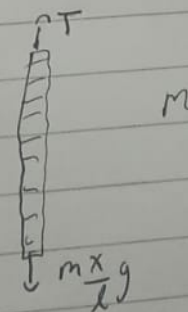

Assumptions :-
1) acceleration of all points in the rope is equal
2) Tension transmitted by the turning point to both sides is equal. 3) Uniform Density

Consider the $l-x$ portion of the rope




$$\therefore m\frac{(l-x)}{l}a = T \Rightarrow a = \frac{T}{m}\left(\frac{l}{l-x}\right)$$

Consider the $x$ portion of the rope




$$\frac{mx}{l}a = \frac{mx}{l}g - T$$

$$a = g - \frac{Tl}{mx}$$

$$g - \frac{Tl}{mx} = \frac{T}{m}\left(\frac{l}{l-x}\right) \Rightarrow \frac{T}{m}\left(\frac{l}{x} + \frac{l}{l-x}\right) = g$$

$$T\left(\frac{-\ell^2}{x(\ell-x)}\right) = mg$$

$$T = mg\left(\frac{x(\ell-x)}{\ell^2}\right)$$

$$a = \frac{T}{m}\cdot\frac{\ell}{(\ell-x)} = g\left(\frac{x}{\ell}\right)$$

Since acceleration of all points in the rope is equal

$$\frac{d^2x}{dt^2} = g\left(\frac{x}{\ell}\right)$$

$$x = C\cdot e^{\sqrt{\frac{g}{\ell}}t}$$

$$x(0) = \ell_0 \Rightarrow x = \ell_0\, e^{\sqrt{\frac{g}{\ell}}t}$$

$$\frac{d^2}{dt^2}(x) = -\frac{g}{\ell}x$$

$$x = c_1 e^{\sqrt{\frac{g}{\ell}}t} + c_2 e^{-\sqrt{\frac{g}{\ell}}t}$$

$x(0) = l_0 \Rightarrow c_1 + c_2 = l_0$

$v(0) = 0 \Rightarrow c_1 - c_2 = 0$

$$v = \sqrt{\frac{g}{\ell}} c_1 e^{\sqrt{\frac{g}{\ell}}t} - \sqrt{\frac{g}{\ell}} c_2 e^{-\sqrt{\frac{g}{\ell}}t}$$

$$c_1 = c_2 = \frac{l_0}{2}$$

$$x = \frac{l_0}{2} e^{\sqrt{\frac{g}{\ell}}t} + \frac{l_0}{2} e^{-\sqrt{\frac{g}{\ell}}t}$$